Part I: Product, Market, and Customer Analysis: Analyze the customized and stock product offerings. This includes: ID the target market segments for each product category; describe both intermediate and final customers; identify and compare the order winners and order qualifiers for each product category; Discuss the implications of product characteristics on supply chain design.

Part II: Supply Chain Analysis: Analyze/conceptualize how their two supply chains work; customized products SC and stock products SC. For each supply chain do the following:

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    - General description that includes at least one graphical representation of each of their SC (see slides 11-14 of Introduction set - Module 1 for a basic idea). Be sure to a) identify the customer order decoupling point; b) identify the push-pull boundary; and c) classify the supply chain using concepts discussed in class (e.g., efficient vs. responsive, lean vs. agile, make-to-stock vs. make-to-order).
    - Forecasting demand. Be sure to discuss a) forecasting approaches that would be appropriate for each supply chain; b) forecasting challenges; and c) expected forecast accuracy and the consequences of forecasting errors.
    - Inventory systems/ reordering policies for raw materials. Include in your analysis in what SC they have the largest number of SKUs, total $, inventory $/item made, and inventory wastes.
    - Supplier management including supplier requirements, number and type of suppliers, supplier relationships, sourcing strategy, and reordering policies.
    - Outbound transportation strategy (from manufacturing to the customers). Include last-mile delivery implications and trade-offs between cost, responsiveness, and customer service.
    - Information technology's role in their SC activities.
    - Identify and justify at least five key performance metrics that management should use to monitor each supply chain.

Given the lack of direct information, make assumptions as necessary.

Part III: Risk and Sustainability Assessment: Analyze Timbuk2's supply chain risks and sustainability elements.

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    - Develop a risk assessment matrix for both supply chains; identify and evaluate at least five major risks, including: supplier disruptions, demand variability, transportation disruptions; inventory shortages; quality problems; geopolitical risks. For each risk: assess probability and impact and propose mitigation strategies.
    - Compare the sustainability implications of the two supply chains. Consider transportation emissions, inventory waste, excess production, material utilization, product lifecycle impacts. Discuss the trade-offs among sustainability, customer service, and cost.

Part IV: Scenario Analysis: Assume demand for customized products increases by 50% over a six-month period following a successful social media campaign. Discuss: Major operational challenges, impact on suppliers, impact on inventory, impact on transportation, impact on customer service, and recommended managerial actions

Part V: Executive Recommendation: Assume Timbuk2's executive team can invest significantly in only one supply chain capability during the next five years. Select one area (for example forecasting and analytics, nearshoring production, digital supply chain technologies) and prepare a recommendation that a) justifies the investment, b) explains expected costs/challenges, c) evaluates impacts on cost, service, flexibility, risk, and sustainability, and d) explains how the recommendation supports competitive advantage

Executive Summary : Begin the report with a one-page Executive Summary written for the Chief Operating Officer (COO) of Timbuk2.